



Effects of calcareous marine algae on milk production, feed intake, energy balance, mineral status, and inflammatory markers in transition dairy cows

E. W. Neville,^{1,2} A. G. Fahey,^{3*} K. G. Meade,³ and F. J. Mulligan¹

¹School of Veterinary Medicine, University College Dublin, Belfield, Dublin 4, Ireland D04 W6F6

²Celtic Sea Minerals, Currabinny, Carrigaline, Co. Cork, Ireland P43 NN62

³School of Agriculture and Food Science, University College Dublin, Belfield, Dublin 4, Ireland D04 R7R0

ABSTRACT

The objective of this experiment was to compare the effects of calcareous marine algae (CMA; Acid Buf, Celtic Sea Minerals) with a limestone-based control on feed intake, milk production, energy balance, serum mineral metabolites, and inflammatory markers in transition dairy cows. Twenty-two multiparous and 10 primiparous cows were assigned to 2 treatments from 25 d before expected parturition until 42 d postpartum. Cows were assigned to treatment according to a randomized complete block design based on parity, pre-experimental body condition score, previous 305-d milk yield, and either fat + protein yield (for multiparous cows) or predicted transmitting ability for milk yield and fat + protein yield (for primiparous cows). Cows were fed a negative dietary cation-anion difference [−50 mEq/kg] total mixed ration (TMR) based on corn silage, grass silage, and straw during the prepartum period and a 50:50 forage:concentrate TMR based on grass silage, corn silage, and concentrate during the postpartum period. The 2 dietary treatments consisted of a control (CON), which contained limestone as the primary calcium source, and CMA, in which limestone was replaced by CMA at 0.42% and 0.47% of dry matter for the pre- and postpartum periods, respectively. The dietary treatments were fed as 2 different concentrate pellets added to the TMR. Cows fed the CMA diet had higher dry matter intake in both the prepartum (+1.08 kg) and postpartum (+0.94 kg) periods compared with cows fed the CON diet. Fat yield (+0.11 kg), fat concentration (+0.43%), and 4% fat-corrected milk (+1.56 kg) were higher in cows fed CMA than in cows fed CON. The concentration of plasma serum amyloid A was reduced and that of serum P was increased on the CMA treatment compared with the CON treatment. These findings demonstrate the benefits of supplementing CMA to dairy cows during the transition period

compared with a CON treatment containing limestone as the primary Ca source.

Key words: calcareous marine algae, dairy cow, dry matter intake, milk production, transition period

INTRODUCTION

The transition period, 21 d prepartum to 21 d postpartum, is one of the most challenging periods for a dairy cow (Grummer, 1995). The “transition” refers to dairy cows going from a near-maintenance state in late gestation and rapidly changing to a state of increased metabolic and nutrient demands required for the onset of lactation (Cardoso et al., 2020). Some of the challenges that occur during the transition period include subclinical hypocalcemia (SCH), SARA, negative energy balance, and reduced immune function and inflammation (Mulligan and Doherty, 2008). Feeding highly fermentable diets in early lactation is necessary to minimize the extent of negative energy balance but it can lead to periods of low ruminal pH and subsequently induce an inflammatory response (Bradford et al., 2015). Low ruminal pH or SARA in early lactation can also reduce DMI (Penner et al., 2007). The calcium requirement of a dairy cow at parturition increases 3-fold while dietary intake is simultaneously reduced by 30%, potentially leading to insufficient blood calcium levels and SCH (McArt and Neves, 2020). Recent research has indicated that inflammation may be partly causal to SCH incidence during the early postpartum period (Horst et al., 2020). A study by Al-Qaisi et al. (2020) revealed how an oral supplement containing soluble calcium and live yeast ameliorated LPS-induced hypocalcemia.

Calcareous marine algae (CMA; Acid Buf or Calmin, Celtic Sea Minerals) is a feed additive produced from *Lithothamnion*, harvested off the coast of Iceland. The CMA supplement contains 95% ash, 30% Ca, 5.5% Mg, 1% P, and 0.7% K. Calcareous marine algae has been used in humans as a Ca and Mg supplement, in research studies and as a commercially available supplement, for many years. Aslam et al. (2010)

Received October 17, 2021.

Accepted April 14, 2022.

*Corresponding author: alan.fahey@ucd.ie

demonstrated that markers of bone mineralization and bone strength were increased in mice supplemented with CMA compared with mice receiving a control treatment containing limestone. Research carried out on CMA in humans has demonstrated improvements in calcium metabolism when supplemented to premenopausal women (Zenk et al., 2018) and reduced pain when supplemented to patients with knee osteoarthritis (Heffernan et al., 2020). Further research, using LPS-treated human macrophage cells, showed CMA to have an anti-inflammatory role in inhibiting nuclear factor kappa B (NF- κ B) activation through a reduction in the phosphorylation and degradation of its upstream inhibitor κ B α , leading to reduced cyclo-oxygenase-2 gene expression (O’Gorman et al., 2012). Calcareous marine algae has also been used extensively in ruminant animal diets, as a ruminal buffer and mineral supplement, both at the farm level and in a research setting (Bernard et al., 2014; Cruywagen et al., 2015; Wu et al., 2015; Neville et al., 2019). Previous research in dairy cows demonstrated that CMA improved ruminal pH regulation and fermentation (Cruywagen et al., 2015), increased milk fat and protein production (Neville et al., 2019), and improved feed efficiency parameters (Bernard et al., 2014). Cruywagen et al. (2015) described CMA as sparingly soluble in water, due to its chemistry and unique physical structure derived from its marine origin, and therefore effective as ruminal buffer while also providing a soluble source of Ca in the rumen. The use of CMA in transition dairy cows is a relatively new concept and could positively affect ruminal fermentation and calcium metabolism through its ability to provide Ca and buffer in the rumen at a time when the risks of SARA and SCH are high. Wu et al. (2015) previously investigated the supplementation of CMA to transition dairy cows. During that experiment, cows were supplemented with either CMA or a control diet prepartum and assigned to a diet containing sodium bicarbonate or CMA postpartum using a 2 $\times$ 2 factorial design. The outcomes of this experiment were higher protein yield for CMA cows versus control cows with no differences detected for DMI or other milk production parameters pre- or postpartum. There has been no previous research carried out on the effects of CMA on inflammatory markers and energy balance in transition dairy cows or the effects of CMA compared with a control containing no ruminal buffer or rumen modifier during the postpartum stage of the transition period.

The objective of this experiment was to compare the effects of supplementing CMA to cows during the transition period on feed intake, milk production, energy balance, serum mineral metabolites, and inflammatory markers compared with a control diet containing lime-

stone as the primary calcium source. Based on previous experiments demonstrating the effects of CMA on ruminal pH (Cruywagen et al., 2015; Neville et al., 2019), we hypothesize that feeding CMA will increase feed intake, and subsequently improve energy balance. We also hypothesize that CMA will reduce the level of inflammatory markers compared with the control diet by avoiding periods of low ruminal pH and potential for LPS to translocate from the rumen to the bloodstream, providing a more soluble supply of Ca in the rumen and inhibiting LPS-induced NF- κ B activation.

MATERIALS AND METHODS

All procedures described in this experiment were approved by the Animal Research Ethics Committee (AREC) at University College Dublin (UCD) and conducted under experimental license from the Health Products Regulatory Authority (HPRA) under the European directive 2010/63/EU and S.I. No. 543 of 2012. Each person who carried out procedures on experimental animals during this experiment was authorized to do so by means of individual authorization from the HPRA.

Experimental Design and Feeding Management

Thirty-two (22 multiparous and 10 primiparous) Holstein cows were selected from the dairy herd at UCD Lyons Farm, Co. Kildare, Ireland. Cows were assigned to treatment according to a randomized complete block design with repeated measures based on parity, pre-experimental BCS (multiparous: range = 2.5–3.75, median = 2.75; primiparous: range = 2.75–3.5, median = 3.25), previous 305-d milk yield ($7,009 \pm 1,404$ kg), and fat and protein yield (556 ± 97 kg) for multiparous cows and PTA for milk yield (255 ± 127 kg), and fat and protein yield (22 ± 7 kg) for primiparous cows. Dry matter intake was the most important variable for the objective of this experiment and considered to have one of the highest levels of variation among the variables that were measured in this experiment. A sample size calculation based on 80% power and a significance level of 0.05 based on postpartum DMI was used to determine sample size for this experiment. A coefficient of variation for postpartum DMI of 18.2% and an expected difference of 18% was estimated by averaging the results of previous studies (Wu et al., 2015; Little et al., 2017; Neville et al., 2019), and it was determined that the sample size for this experiment was $n = 16$.

The 2 dietary treatments consisted of a control diet (CON) and a CMA diet (CMA), in which limestone was replaced by CMA at 0.42% and 0.47% DM for the

Table 1. Ingredient composition of the control and calcareous marine algae diets during prepartum and postpartum periods

Ingredient (% of DM)	Dietary treatment ¹			
	Prepartum		Postpartum	
	CON	CMA	CON	CMA
Corn silage	43.56	43.56	25.7	25.7
Straw (barley)	22.29	22.29	—	—
Grass silage	18.31	18.31	25.7	25.7
Soybean meal (48% CP)	7.43	7.43	10.72	10.72
Wheat grain (finely ground)	—	—	9.48	9.49
Barley grain (finely ground)	—	—	8.00	8.04
Parlor concentrate ²	—	—	7.94	7.94
Soychlor ³	6.76	6.76	—	—
Corn grain (finely ground)	—	—	4.54	4.54
Molasses (sugar cane)	—	—	3.28	3.28
Soy hulls	—	—	2.47	2.47
Dry cow premix ⁴	1.22	1.22	—	—
Milking cow premix ⁵	—	—	0.23	0.23
White salt (NaCl)	—	—	0.47	0.47
Mono-dicalcium phosphate	—	—	0.47	0.47
Vegetable oil	—	—	0.23	0.23
Limestone	0.34	0.00	0.61	0.14
Calcined magnesite	0.04	0.00	0.14	0.09
Calcareous marine algae ⁶	0.00	0.43	0.00	0.47

¹Treatments: CON = control; CMA = calcareous marine algae (95% ash, 30% Ca, 5.5% Mg, 1% P, and 0.7% K).

²Parlor concentrate ingredients (DM basis): 22.5% barley grain, 22.5% corn grain, 10% corn distillers dried grains with solubles, 9% sugar beet pulp, 21% soybean meal, 2.5% soy hulls, 1.5% palm oil, 4.5% sugar cane molasses, 1.6% mono-dicalcium phosphate, 1.6% limestone, 1.8% salt, 1.5% MgO.

³Anionic salt supplement (Landus Cooperative); contained −2,980 mEq of DCAD/kg, 20.1% CP, 0.99 Mcal/kg, 4.54% Ca, 0.53% P, and 2.84% Mg.

⁴Dry cow premix contained 24% corn grain (carrier), 1% Ca, 10% P, 12% Mg, 4% Na, 6% Cl, 2,767 mg/kg of Mn, 5,000 mg/kg of Zn, 2,000 mg/kg of Cu, 33 mg/kg of Se, 333 mg/kg of I, 67 mg/kg of Co, 134 mg/kg of biotin, 500 IU/kg of vitamin A, 167 IU/kg of vitamin D3, and 8,000 IU/kg of vitamin E.

⁵Milking cow premix contained 30% Ca, 50 mg/kg of Co, 10,400 mg/kg of Cu, 390 mg/kg of I, 130 mg/kg of Se, 21,600 mg/kg of Mn, 32,500 mg/kg of Zn, 10,000 IU of vitamin A, 3,000 IU of vitamin D3, 13 IU of vitamin E.

⁶*Lithothamnion calcareum*.

pre- and postpartum periods, respectively. The dietary treatments were fed as 2 different concentrate pellets added to the TMR. The ingredient composition of the 2 dietary treatments (pre- and postpartum) are outlined in Table 1. The inclusion rate of the CMA additive was based on a predicted daily DMI of 11.7 kg of DM prepartum and 21.4 kg of DM postpartum to ensure daily intakes of 50 g and 100 g of CMA per cow per day in the pre- and postpartum periods, respectively. The dosage rates of CMA were determined from previous experiments (Wu et al., 2015). Limestone and MgO levels were adjusted in the CMA treatment to balance diets for Ca and Mg.

The diets were formulated using recommendations from INRA (2018). Prepartum diets were designed to

supply 100% of the energy requirements of a 650-kg nonlactating dairy cow in late gestation with a predicted DMI of 11.7 kg of DM/cow per day. Both prepartum diets were formulated to contain 15% starch and water-soluble carbohydrates (WSC), 0.9% Ca, 0.46% Mg, and DCAD of −53 mEq/kg, based on corn silage, grass silage, barley straw, and soybean meal. Postpartum diets were designed to supply 95% of the energy requirements of a 650-kg lactating dairy cow yielding 32 kg of milk/d containing 3.9% fat and 3.3% protein with a predicted DMI of 21.4 kg of DM/cow per day and a feed allocation of 22.5 kg of DM/cow per day, to allow for 5% refusals. The 2 postpartum diets were formulated to contain 49% concentrate, 29% starch and WSC, and 22% NDF from forage, based on grass silage, corn silage, and concentrate. Multiparous and primiparous cows were trained to use the computerized feeding stations before the experiment. All cows were introduced to the feeding stations and fed a control diet from 28 d before expected calving date as an acclimatization to the diet, feeding system, and the barn. The experimental diets were introduced to the cows from 25 d prepartum.

Cows received their complete prepartum diet and most of their postpartum diet through a TMR once daily at 0900 h. Each treatment was mixed separately using a Keenan Feeder (Keenan Feeding Systems) and each dietary treatment received the same total mixing time, 10 min after all the ingredients were added to the mixer wagon. During milking, cows received a pellet (1 kg as fed at a.m. milking and 1 kg as fed at p.m. milking) to incentivize them into the milking parlor. Parlor troughs were inspected before and after each cow entered the parlor and any refusals were recorded. Water was available ad libitum and water troughs were cleaned daily.

The grass silage used consisted predominantly of perennial ryegrass (*Lolium perenne*). The crop was cut using a mower-conditioner during the early boot stage of vegetation (growth stage 41; Zadoks et al., 1974), wilted for 16 h, and harvested with a Krone Big X (Krone GmbH and Co.) forage harvester (mean particle length 50 mm). The crop was then ensiled under a black polythene cover without the use of an additive. The corn (*Zea mays*, variety Tekni) used for corn silage was grown with the aid of plastic film (Samco Agricultural Manufacturing Ltd.). The crop was harvested at the dough stage (growth stage 85; Lancashire et al., 1991) using a Krone Big X (Krone GmbH and Co.) precision chop forage harvester (mean particle length 25 mm). The harvester was equipped with a kernel processor to improve starch digestibility. The harvested corn silage was ensiled under a black polythene cover without the use of an additive.

Before the experiment, fresh samples of both grass and corn silage were sent to a forage laboratory (Trouw Nutrition, Belfast, UK) for near infrared analysis (FOSS NIR systems 5000) to provide nutrient values so that the basal TMR could be formulated. The ingredient composition of pre- and postpartum diets is presented in Table 1. The concentrate portion of the postpartum diet was fed in the form of a pellet and added to the TMR to aid in mixing accuracy. The concentrate pellet used in the TMR was manufactured by Brett Brothers Ltd. The milking parlor pellet was manufactured by Gain Feeds.

Animal Care and Housing

Each cow was on the experiment for a minimum of 57 d (57–67 d), from 25 d before expected calving date to 42 d postpartum. Sampling and data collection were carried out from 21 d prepartum to 42 d postpartum. Cows were housed in a freestall barn with 1.25 stalls per cow available during the pre- and postpartum period. When cows were observed to be within 24 h of calving, they were moved to a loose pen to facilitate calving while still receiving their respective diets. Cows had ad libitum access to the TMR for 22 h/d through specific computerized feeding boxes (RIC System, In-sentec B.V.). Twenty feeding boxes were available for the 32 cows enrolled in this experiment (0.6 boxes/cow), with boxes divided evenly across treatments. Cows were milked twice daily, at 0800 and 1600 h, in a rotary milking parlor (Dairymaster), and stalls were cleaned and new sawdust bedding added twice daily.

Data Collection, Sampling Procedures, and Sample Analyses

Samples of TMR and refusals were collected daily and dried at 104°C in a forced air oven for 16 h to determine the DM content of the TMR and refusals. Individual daily feed intakes were recorded in the computerized feeding system and used, in combination with daily TMR DM content, to calculate DMI. Daily TMR allocation was 105% of the DMI from the previous day. Samples of TMR were taken from each treatment during feed-out 3 times per week, pooled into weekly samples by treatment, and stored at –20°C until analysis. Concentrate, grass silage, corn silage, and straw samples were collected weekly and stored at –20°C before analysis. Subsamples of the composite TMR from both pre- and postpartum periods, corn silage, grass silage, barley straw, soybean meal, and concentrate samples were dried at 55°C for 72 h. The subsequent dried samples were ground using a Nor-

ris hammer mill fitted with a 1-mm screen (Lab Mill, Christy Turner). Ash content was determined by incineration of a 5-g sample in a muffle furnace (Nabertherm GmbH) at 550°C for 5.5 h. The N content of the feed was determined by combustion using a Leco 528 instrument (Leco Instruments UK). Crude protein was then calculated using $N \times 6.25$. Neutral detergent fiber and ADF were determined according to the method of Van Soest et al. (1991) using the Ankom 220 Fiber Analyzer (Ankom Technology). As part of the NDF procedure, both sodium sulfite (Na_2SO_3) and heat-stable α -amylase (both from Ankom Technology) were used for the analysis of TMR, corn silage, barley straw, and concentrate subsamples, whereas only Na_2SO_3 was used for grass silage subsamples. Starch content was determined using the Megazyme Total Starch Assay Procedure (product no. K-TSTA; Megazyme International Ltd.). Briefly, 0.2 mL of aqueous ethanol was added to 100 mg of sample and mixed using a vortex. Then, 2 mL of dimethyl sulfoxide was added and the sample incubated in a boiling water bath for 5 min. As the samples were removed from the boiling water, 3 mL of thermostable α -amylase (50 mM, pH 7.0) plus calcium chloride (5 mM) was added, followed by a further incubation step in a boiling water bath for 6 min, vortexing vigorously after 2, 4, and 6 min. After 4 mL of sodium acetate buffer (200 mM, pH 4.5) plus calcium chloride (5 mM) and amyloglucosidase (0.1 mL, 3,300 U/mL) were added, samples were incubated at 50°C for 30 min. Then, 2.0 mL of sample was centrifuged at $19,000 \times g$ for 5 min, followed by mixing 0.1-mL aliquots of each sample with 3.0 mL of glucose determination reagent and incubated at 50°C for 20 min. Absorbance of each sample was then measured at 510 nm on a spectrophotometer. Starch content was determined using the following equation (AOAC International, 2005):

$$\begin{aligned} \text{Starch, \%} &= \text{absorbance of sample} \times \text{conversion factor} \\ &\text{for absorbance to } \mu\text{g of glucose} \times \text{extraction volume} \\ &\times (\text{sample dilution factor/sample weight}) \times 0.90. \end{aligned}$$

The method outlined by Birch and Mwangiwa (1974) was used for the analysis of WSC. Ether extract was measured using a Soxtec instrument (Tecator) according to method 107 (AOAC, 1970).

Milk samples were collected on 2 separate days each week during the postpartum period. Each day consisted of an a.m. and p.m. sample pooled in proportion to the specific a.m. and p.m. yields to create 1 milk sample/d and 2 milk samples/cow for each week. Samples were preserved (Broad Spectrum Microtabs II, D&F Con-

trol Systems Inc.) and stored at 4°C until analyzed. Milk samples were analyzed within 48 h of collection. Daily milk yields were automatically recorded using a Weighall milk meter system (Dairymaster). Concentrations of milk fat, protein, lactose, urea, and casein were determined by a commercial milk laboratory (Progressive Genetics, Enfield, Ireland) using infrared analysis (CombiFoss 5000, Foss Analytical). Values for ECM and 4% FCM were calculated using the following formulas (Tyrrell and Reid, 1965, and Gaines, 1928, respectively):

$$\text{ECM} = (0.3273 \times \text{milk yield kg}) + (7.65 \times \text{milk protein kg}) + (12.97 \times \text{milk fat kg});$$

$$4\% \text{ FCM} = (0.4 \times \text{milk yield kg}) + (15 \times \text{fat yield kg}).$$

Energy balance was calculated as daily mean energy balance [unit of energy for lactation, equivalent to 1,700 kcal of NE_L (unité fourragère lait, **UFL**)/d] and area under the curve (**AUC**) energy balance (UFL × wk). Total AUC for energy balance over the 6-wk postpartum period was calculated using the trapezoidal rule (Cardoso et al., 2011). Body condition score of each cow was assessed once per week prepartum and postpartum and at parturition, by the same trained individual using the 5-point scale method, as described by Edmonson et al. (1989). Body condition score was analyzed as a continuous variable. The change in BCS over the postpartum period was reported, in addition to the weekly BCS for each cow. The BCS change from wk 1 to 3 was calculated as the difference in BCS between wk 1 and 3. Similarly, BCS change from wk 4 to 6 was calculated as the difference in BCS from wk 4 to 6. The BW of each cow was recorded daily during the postpartum period after the a.m. milking and before access to feed.

Blood samples were harvested by jugular venipuncture on d -7, at <6 h, 24 h, 48 h, and d 7, 14, 21, and 28 relative to parturition. Samples collected on d -7, 7, 14, 21, and 28 were collected 6 h after feed delivery. Blood samples were collected into evacuated tubes (Becton Dickinson), which were coated with a clot activator, fluoride oxalate, or lithium heparin. Upon collection, tubes were placed on ice until processed accordingly. Samples from the tubes with clot activator were stored in 4°C for 12 h, centrifuged at 2,100 × *g* and 4°C for 20 min to obtain serum, and used for the determination of Ca, P, Na, K, Cl, Mg, nonesterified fatty acid (**NEFA**), and BHB concentrations. Fluoride oxalate blood tubes were immediately centrifuged at 1,800 × *g* and 4°C for

15 min to obtain plasma for glucose determination. Lithium heparin tubes were immediately centrifuged at 2,100 × *g* and 4°C for 10 min to obtain plasma for serum amyloid A (**SAA**) and haptoglobin (**Hp**) determinations. Blood samples were analyzed for Ca, P, Na, K, Cl, Mg, NEFA, BHB, and glucose at the Veterinary Clinical Pathology Laboratory (School of Veterinary Medicine, UCD, Dublin, Ireland) using Randox kits according to the manufacturer's instructions, with an RX IMOLA analyzer (Randox Laboratories). Serum amyloid A and Hp were measured using ELISA kits (Tridelta Diagnostics Ltd.) as described by Gozho et al. (2007).

Data Screening and Statistical Analyses

Data residuals were examined for normality using the UNIVARIATE procedure of SAS (version 9.4, SAS Institute Inc.). Following assessment of normality, outliers [± 3 standard deviations (**SD**) from the mean] were removed. Data for milk yield contained 8 observations that were outside ± 3 SD from the mean, which were removed from the data set (5 time points for 5 different CON cows and 3 time points for 3 different CMA cows). No outliers were removed for any of the other variables. Milk production, intake, energy balance, BCS, BW, and blood data were analyzed as repeated measures using the MIXED procedure (SAS, version 9.4). The model included fixed effects of treatment, week, block, and parity, as well as treatment by week interaction, with cow considered the random effect. Repeated measures were modeled for week, and variance-covariance structures were selected based on the lowest Bayesian information criterion. Milk fat concentration, DMI, and Na, K, Cl, and BHB serum concentrations required a compound symmetry covariance structure. Serum Mg concentration required the first-order autoregressive covariance heterogeneous structure, and SAA required the Toeplitz covariance structure. All other milk, BCS, BW, plasma, and serum data required the first-order autoregressive covariance structure. All data presented in Tables 3 to 6 are expressed as least squares means $\pm$ standard error of the means (**LSM** $\pm$ **SEM**). Statistical significance was declared at $P < 0.05$.

RESULTS

Chemical Analysis of TMR

The nutrient compositions of the CON and CMA treatments in the prepartum and postpartum periods are presented in Table 2. Values for protein truly di-

Table 2. The analyzed and predicted chemical and nutrient profile of the control and calcareous marine algae diets during the prepartum and postpartum periods

Item	Dietary treatment ¹			
	Prepartum		Postpartum	
	CON	CMA	CON	CMA
Chemical composition (% of DM)				
DM	40.7	40.1	40.0	40.6
CP	13.0	13.1	16.4	15.9
PDIN ²	8.11	8.11	10.85	10.85
PDIE ²	7.39	7.39	10.42	10.42
NDF	44.2	46.3	29.2	28.0
Forage NDF ³	93.16	93.16	25.4	25.4
ADF	29.2	29.2	17.2	17.2
Ash	9.81	9.70	7.98	7.88
Starch	14.9	14.5	24.8	24.9
Sugar	4.3	4.0	4.08	4.36
Ca	0.87	0.87	0.83	0.84
P	0.35	0.35	0.45	0.46
Mg	0.46	0.46	0.32	0.33
K	1.80	1.80	1.75	1.76
Na	0.08	0.08	0.30	0.31
Cl	1.50	1.50	0.85	0.85
S	0.20	0.20	0.18	0.18
DCAD (mEq/kg)	-52.2	-52.2	226	233
ME ² (MJ/kg)	9.91	9.91	12.0	12.0
Net energy ² (UFL/kg)	0.76	0.76	0.99	0.99
NE _L ⁴ (Mcal/kg of DM)	1.33	1.33	1.74	1.74

¹Treatments: CON = control; CMA = calcareous marine algae.

²As calculated by using INRatio 4.07 feed formulations program, based on ingredient analyses, and similar for all treatments. PDIN = protein truly digestible in the small intestine when nitrogen is limiting microbial protein synthesis; PDIE = protein digested in the small intestine when energy is limiting microbial protein synthesis; UFL (unité fourragère lait) = unit of energy for lactation.

³Contribution of the forage component of the diet to NDF.

⁴NE_L = net energy for lactation at production level (NRC, 2001).

gestible in the small intestine when nitrogen is limiting microbial protein synthesis (PDIN), protein digested in the small intestine when energy is limiting microbial protein synthesis (PDIE), UFL, and NE_L were predicted based on the INRatio 4.07 computer program (INRA, 2018) and NRC (2001).

Prepartum Variables

Table 3 presents the effects of the CMA and CON treatments on DMI, BCS, serum energy metabolites, and serum mineral concentration during the prepartum period. The CMA treatment increased DMI ($P = 0.03$) and reduced serum NEFA concentration ($P = 0.10$) compared with the CON treatment, whereas no differences in serum BHB concentration or BCS were observed between the 2 groups during the prepartum period. We found no differences between treatments for serum concentrations of Ca, Mg, P, Na, K, and Cl, and no treatment by week interactions were detected for any of the prepartum variables.

Postpartum Performance

Table 4 shows the effect of CON and CMA on DMI, energy balance, BCS, BW, and serum and plasma energy metabolites. Cows fed CMA had higher DMI than cows fed CON ($P = 0.05$) during the postpartum period. Mean energy balance was similar for both treatments, whereas AUC energy balance (UFL $\times$ wk) was higher on the CMA treatment ($P = 0.06$) than on the CON treatment. Mean BCS and BCS change from 1 to 3 wk and from 4 to 6 wk were similar between treatments. Mean BW and days to nadir BW were not affected by treatment. Concentrations of BHB and NEFA in serum and glucose in plasma were similar between the CMA and CON treatments. Nadir BW (CON: 584 kg, CMA: 594 kg) and days to nadir BW (CON: 12.7 d, CMA: 12.2 d) did not differ between treatments (data not shown).

The effects of treatments on milk production during the postpartum period are outlined in Table 5. Milk yield, protein yield, and fat + protein yield were similar

Table 3. The effect of control and calcareous marine algae on DMI, BCS, serum energy metabolites, and serum mineral concentration during the prepartum period

Variable	Treatment ¹		SEM	P-value ²
	CON	CMA		
DMI, kg/d	12.23	13.31	0.39	0.03
BCS	2.99	3.01	0.10	0.82
Serum energy metabolite (mM)				
NEFA ³	0.22	0.11	0.05	0.10
BHB	0.52	0.56	0.04	0.46
Serum mineral concentration (mM)				
Ca	2.28	2.27	0.02	0.72
Mg	0.86	0.86	0.02	0.85
P	2.18	2.20	0.08	0.85
Na	145.0	145.8	0.67	0.44
K	5.33	5.41	0.16	0.70
Cl	106.7	107.7	1.02	0.52

¹Treatments: CON = control; CMA = calcareous marine algae.

²P-value for treatment effect. No significant treatment × week interactions were detected for any of the above variables.

³NEFA = nonesterified fatty acid.

between treatments. The CMA treatment increased 4% FCM ($P = 0.03$) and fat yield ($P = 0.04$) compared with the CON treatment. The yield of ECM was higher on the CMA treatment than on the CON treatment ($P = 0.07$). Cows fed CMA had higher milk fat concentrations than cows fed CON ($P < 0.01$). There was no difference between treatments for milk protein, lactose, casein, or urea concentrations.

The effect of CON and CMA treatments on postpartum serum mineral concentrations plasma inflammatory markers are shown in Table 6. No differences were detected between treatments for concentrations of serum Ca, Mg, Na, K, or Cl, whereas the CMA treatment

increased the serum concentration of P ($P < 0.01$) compared with the CON treatment during the postpartum period. Plasma SAA was lower in cows fed CMA ($P = 0.07$) than in cows fed CON, but Hp was not affected by treatment. No treatment by week interactions were detected for any of the postpartum variables.

DISCUSSION

Prepartum Variables

Cows fed the CMA diet had similar serum concentrations of Ca, P, and Mg compared with cows fed the

Table 4. The effect of control and calcareous marine algae on DMI, energy balance, BCS, BW, and energy metabolites during the postpartum period

Variable	Treatment ¹			P-value ²	
	CON	CMA	SEM	Trt	Wk
DMI (kg/d)	19.14	20.08	0.39	0.05	<0.01
Energy balance					
Mean (UFL/d) ³	0.02	0.55	0.44	0.40	<0.01
AUC (UFL × wk) ³	0.69	3.51	1.06	0.06	—
BCS and BW					
Mean BCS	2.79	2.81	0.08	0.65	0.72
BCS change wk 1–3	−0.06	−0.05	0.05	0.85	—
BCS change wk 4–6	0.12	0.08	0.05	0.62	—
Mean BW, kg	598.0	620.5	16.32	0.33	<0.01
Serum and plasma energy metabolite (mM)					
BHB	0.60	0.63	0.02	0.31	<0.01
NEFA ⁴	0.27	0.25	0.03	0.64	<0.01
Glucose	3.24	3.20	0.04	0.54	<0.01

¹Treatments: CON = control; CMA = calcareous marine algae.

²Trt = P-value for treatment effect. Wk = P-value for week effect. No significant treatment × week interactions were detected for any of the above variables.

³UFL (unité fourragère lait) = unit of energy for lactation. AUC = area under the curve.

⁴NEFA = nonesterified fatty acid.

Table 5. The effect of control and calcareous marine algae on milk production during the postpartum period

Variable	Treatment ¹			P-value ²	
	CON	CMA	SEM	Trt	Wk
Output (kg/d)					
Milk yield	31.04	30.71	0.76	0.75	<0.01
4% FCM ³	30.09	31.65	0.52	0.03	<0.01
ECM ⁴	34.45	35.76	0.53	0.07	<0.01
Fat yield	1.18	1.29	0.04	0.03	0.15
Protein yield	1.16	1.17	0.02	0.96	<0.01
Fat + protein yield	2.33	2.44	0.06	0.16	<0.01
Composition (%)					
Fat	3.81	4.24	0.09	<0.01	<0.05
Protein	3.72	3.82	0.05	0.17	<0.01
Lactose	4.43	4.42	0.02	0.81	<0.01
Casein	2.87	2.95	0.05	0.21	<0.01
Urea (mg/dL)	14.00	14.33	0.77	0.74	0.71

¹Treatments: CON = control; CMA = calcareous marine algae.

²Trt = *P*-value for treatment effect; Wk = *P*-value for week effect. No significant treatment × week interactions were detected for any of the above variables.

³Calculated using the equation $(0.4 \times \text{milk yield kg}) + (15 \times \text{fat yield kg})$.

⁴Calculated using the equation $(0.3273 \times \text{milk yield kg}) + (7.65 \times \text{milk protein kg}) + (12.97 \times \text{milk fat kg})$.

CON diet during the prepartum period. This agrees with previous work by Wu et al. (2015). Mineral-related challenges in dairy cows typically occur at the onset of or in the days following parturition (Goff, 2008). Therefore, differences in Ca and Mg supply between the CMA and CON treatments due to potential differences in bioavailability may not be detected prepartum because the cows' mineral metabolism is not challenged to the same degree as it is in the days following parturition. Furthermore, serum mineral concentrations were all within the expected range for healthy prepartum dairy cows (Quiroz-Rocha et al., 2009).

The higher DMI observed with the CMA treatment during the prepartum period disagrees with previous work by Wu et al. (2015). Increased DMI might be related to improvements in ruminal pH and fermentation and a faster rate of passage from the rumen due to increased digestibility of nutrients. Calcium supply and status are important for rumen contractions in the early postpartum and will partly dictate how much DM can be consumed by the cow (Goff, 2008). Although we found no differences in major mineral concentrations prepartum, increased availability of Ca and Mg from the CMA treatment may have contributed to these differences in DMI prepartum. However, further research is required to investigate this.

The CMA-fed cows had lower concentrations of serum NEFA postpartum compared with CON-fed cows during this study. Excess serum concentrations of NEFA in dairy cows is commonly used as indicator of poor metabolic status and negative energy balance during the transition period (Mulligan and Doherty, 2008). Increased DMI and subsequent improvements in energy balance with CMA during the prepartum period may have reduced NEFA concentration. However, these results must be taken in context because the level of serum NEFA detected in CON cows was well within normal levels and not indicative of energy balance issues for prepartum dairy cows. The similarity between treatments for serum BHB concentration was in line with results of previous research (Wu et al., 2015).

Postpartum Performance

Increased DMI with CMA compared with CON during the prepartum period of this experiment continued into the postpartum period. These results differ from that of Wu et al. (2015), who detected no differences in

Table 6. The effect of control and calcareous marine algae on serum mineral and plasma acute phase protein concentrations during the postpartum period

Variable	Treatment ¹			P-value ²	
	CON	CMA	SEM	Trt	Wk
Serum mineral (mM)					
Ca	2.12	2.15	0.03	0.46	<0.01
Mg	0.92	0.92	0.02	0.88	<0.01
P	1.93	2.13	0.06	0.01	<0.01
Na	143.9	143.9	0.21	0.85	<0.01
K	5.12	5.07	0.12	0.79	<0.01
Cl	103.2	102.5	0.32	0.14	<0.01
Acute phase protein					
Serum amyloid A (µg/mL)	73.2	57.8	6.24	0.07	0.01
Haptoglobin (mg/mL)	0.78	0.70	0.19	0.85	<0.01

¹Treatments: CON = control; CMA = calcareous marine algae.

²Trt = *P*-value for treatment effect; Wk = *P*-value for week effect. No significant treatment × week interactions were detected for any of the above variables.

DMI with CMA during the postpartum period. However, the CMA versus CON comparison in our experiment and in the Wu et al. (2015) experiment need to be considered in the context of the basal diet used. The control diet used by Wu et al. (2015) in the postpartum period contained 1.26% sodium bicarbonate. The use of sodium bicarbonate has been shown to increase DMI compared with control diets (Kawas et al., 2007) and compared with diets containing CMA in mid lactation (Neville et al., 2019; Cruywagen et al., 2015). Therefore, the diets used by Wu et al. (2015) did not allow for an accurate comparison of CMA versus limestone only on DMI during the postpartum period. Changes in mineral status are difficult to relate to DMI differences in our experiment because we detected no differences in the major serum mineral concentrations except for serum P. Reduced DMI is commonly cited as a negative consequence of low ruminal pH and disrupted fermentation (Kleen et al., 2003). Therefore, the evidence supporting the effect of CMA on ruminal fermentation within the literature (Cruywagen et al., 2015; Neville et al., 2019) is a more likely cause of the DMI increases compared with the CON treatment. The reduction in plasma SAA concentration with CMA-fed cows discovered during this experiment could also contribute to differences in DMI between the CMA and CON treatments. Links between inflammation, SARA, and DMI in early lactation have been discussed extensively in recent years (Bradford et al., 2015). The release of inflammatory mediators during an immune response can initiate a reduction in DMI around calving, because of their potent anorexic effects (Horst et al., 2021). Therefore, the tendency for differences in inflammatory mediators between CON-fed and CMA-fed cows could be indicative of greater inflammation experienced by CON-fed cows and may have contributed to the reduced DMI seen in the CON cows during this study.

Results of this experiment indicate that cows supplemented with CMA during the transition period can maintain better energy balance, measured as AUC energy balance, compared with cows fed the CON treatment. The effects of CMA on energy balance in transition dairy cows has not been measured previously. Increased energy balance in cows during early lactation can have far-reaching consequences later in lactation, such as better fertility performance (De Vries and Veerkamp, 2000). The increased energy balance on the CMA treatment was caused by greater DMI throughout the prepartum and postpartum periods. These findings indicate that not all of the increase in DMI in the CMA-fed cows was apportioned toward increased FCM production; improved energy balance for cows fed CMA was also evident in this experiment. The improvements in energy balance did not translate

into improvements in BCS, BW, or serum concentrations of energy metabolites during the postpartum period. However, such improvements in energy balance are subtle and may not be detectable in BCS, BW, or serum energy metabolite changes over the duration of this study (6 wk).

Milk production during the postpartum stage of this study was influenced by the CMA treatment. Although we detected no differences for milk yield, yields of 4% FCM and ECM were higher in cows fed the CMA diet than in cows fed the CON diet. Previous work by Wu et al. (2015) reported no effects of the CMA treatment on milk, FCM, or ECM yield during the early postpartum period compared with the control treatment. However, the inclusion of sodium bicarbonate in the control diet used by Wu et al. (2015) may have influenced milk production variables. Previous work investigating the effects of CMA in mid-lactation dairy cows, using similar diets to those of the current study, also found no differences in milk yield, FCM, or ECM (Neville et al., 2019). The improvements in both FCM and ECM detected in the current study but not in previous work are likely explained by a combination of diet type and stage of lactation. The lack of effect of CMA on milk yield is consistent with previous work (Bernard et al., 2014; Neville et al., 2019).

Milk fat concentration and fat yield were higher on the CMA treatment compared with the CON treatment. The difference in milk fat concentration of 0.43% in favor of CMA is highly significant ($P < 0.01$) and very large from a biological perspective. Both Cruywagen et al. (2015) and Neville et al. (2019) reported similar effects of CMA on milk fat concentrations, albeit with cows later in lactation. However, Wu et al. (2015) found no effect of CMA on milk fat concentration or yield. The inclusion of sodium bicarbonate in the control diet used by Wu et al. (2015) would have removed any potential difference in milk fat production between the CMA and CON treatments. In a previous experiment carried out by our research group (Neville et al., 2019), we detected no differences in fat concentration or fat yield between the CMA treatment and the sodium bicarbonate treatment. Milk fat synthesis is commonly associated with ruminal fermentation (Allen, 1997) and whether biohydrogenation pathways for fatty acids in the rumen are complete or not (Fuentes et al., 2009). Although ruminal pH and ruminal fermentation variables were not measured in the current study, there is consistent evidence from previous studies that CMA has a positive effect on ruminal pH and VFA production. Our previous research (Neville et al., 2019) is the most comparable research to the current experiment, in terms of diet composition, environment, and ingredients used. Results from that work showed

increased mean ruminal pH and reduced time spent below pH 5.5, in addition to the increase in milk fat production. Therefore, improved rumen fermentation and more complete biohydrogenation pathways, related to increased ruminal pH, is the most likely mechanism underlying the improvements in milk fat concentration and yield.

Protein concentration, protein yield, and fat + protein yield were not affected by the inclusion of CMA during this experiment. Similar findings were reported by Wu et al. (2015), Cruywagen et al. (2015), and Bernard et al. (2014) for milk protein concentration. Results of this experiment disagree with previous work in our research group (Neville et al., 2019) and Cruywagen et al. (2015); both of these studies noted improvements in protein yield and fat + protein yield with the CMA treatment compared with a control treatment. Our previous work (Neville et al., 2019) also reported an increase in protein concentration with cows fed CMA compared with cows fed sodium bicarbonate and an increase in protein concentration with CMA plus marine MgO compared with the control treatment. Wu et al. (2015) found that CMA supplemented during the prepartum period increased protein yield postpartum, which also differs from the results of the current experiment.

The CMA treatment had no effect on postpartum serum Ca, Mg, Na, K, or Cl concentrations compared with the CON treatment. This agrees with work carried out by Wu et al. (2015). Calcareous marine algae provides a source of Ca and Mg in a different physical and chemical form from typical Ca sources such as limestone and typical Mg sources such as MgO. In contrast to the other serum mineral concentrations measured during this study, serum P concentration was increased on the CMA treatment during the postpartum period. This finding is not consistent with the results of Wu et al. (2015). Leno et al. (2017), investigating Ca-Mg dolomite as an alternative to limestone and MgO, reported similar results to our study. They observed an increase in plasma P concentration and no effect on plasma Ca concentration during the postpartum period. Phosphorus is a major component of bone mineral, second only to Ca (Goff, 1999). Phosphorus concentrations are closely related to plasma PO_4 concentrations, which are indirectly regulated by parathyroid hormone (PTH) and a calcium negative feedback loop (Lean et al., 2006). Calcareous marine algae provides minimal amounts (1 g/cow per day) of supplemental P, which is not likely to affect serum P concentrations. Cruywagen et al. (2015) described CMA as sparingly soluble in water due to its chemistry, fine particle size, and unique physical form. This would indicate that CMA is solubilized in rumen conditions and its mineral contents

(Ca and Mg) released into rumen fluid. The carryover effects on the absorption of Ca and Mg from the rumen may have affected Ca metabolism and resulted in higher serum P concentrations in cows supplemented with CMA. Dairy cows secrete PTH during periods of low serum Ca, consequently stimulating renal conservation of Ca and the release of Ca from bone stores (Goff et al., 1986). Therefore, potential differences in dietary Ca absorption between CMA and CON could have been masked by PTH secretion, explaining why there was no effect of CMA on serum Ca concentration to coincide with the increase in serum P concentration. In monogastrics, limestone can have antagonistic effects on dietary P absorption during the digestion process (Li et al., 2021). The difference in Ca sources used during this study—CMA instead of limestone—may have affected the availability of P for absorption in the lower digestive tract and hence differences in serum P concentration. These findings will need to be investigated further to determine mechanisms behind the effect of CMA on serum P concentration.

The results of this experiment agree in part with our hypothesis because plasma SAA concentration was reduced but plasma Hp concentration was unaffected by the CMA treatment compared with the CON treatment. No previous experiments have reported the effects of CMA on SAA and Hp in dairy cows. The positive effects of CMA on ruminal pH (Cruywagen et al., 2015; Neville et al., 2019) and the findings of O’Gorman et al. (2012), where CMA had an anti-inflammatory effect in rodents, contributed to the hypothesis of this study that CMA would reduce the concentration of inflammatory markers SAA and Hp in the early postpartum period. Ruminal fermentation has been cited as a common precursor of diet-induced inflammation (Zebeli et al., 2012), and different types of SARA can determine the extent of dietary-induced inflammation experienced by the dairy cow. The source of supplemental Ca offered to transition dairy cows may be very important because Ca status is pivotal to the functionality of the cow’s immune system (Martinez et al., 2014). A recent study by Al-Qaisi et al. (2020) reported that an oral supplement containing soluble Ca and live yeast lessened the extent of LPS-induced hypocalcemia, with authors citing the symbiotic effects of the live yeast’s improvement in rumen conditions and the oral Ca effects on Ca metabolism to explain these findings. The reduction in plasma SAA reported in the current experiment are likely explained by improvements in ruminal pH and fermentation with the CMA treatment. The potential increased supply of soluble Ca in the rumen provided by the CMA treatment will need to be investigated further due to the absence of treatment differences for serum Ca concentration.

CONCLUSIONS

Calcareous marine algae supplemented to transition dairy cows increased DMI both pre- and postpartum as well as some important milk production variables such as ECM yield and milk fat concentration compared with cows fed the CON diet, which contained limestone as the primary Ca source. Estimated energy balance postpartum was subsequently improved with the CMA treatment as a result of increased DMI. The concentration of plasma SAA was reduced in the CMA treatment postpartum, indicating a link between CMA, inflammation, and feed intake around parturition. This phenomenon warrants further investigation. Other blood variables such as prepartum NEFA concentration and postpartum P concentration were also positively affected by the addition of CMA to the diet. These results offer dairy producers and nutritionists new information on the implications of adding CMA to transition cow programs and a possible tool to improve the health and production of dairy cows during this critical period.

ACKNOWLEDGMENTS

This experiment was co-funded by the European Regional Development Fund (ERDF) under Ireland's European Structural and Investment Funds Programmes 2014-2020. This funding was through an Enterprise Ireland (Dublin, Ireland) Innovation Partnership with Marigot Ltd. (Cork, Ireland). The authors thank the students and staff at University College Dublin who assisted during this experiment. The authors have not stated any conflicts of interest.

REFERENCES

- Al-Qaisi, M., S. K. Kvidera, E. A. Horst, C. S. McCarthy, E. J. Mayorga, M. A. Abeyta, B. M. Goetz, N. C. Upah, D. M. McKilligan, H. A. Ramirez-Ramirez, L. L. Timms, and L. H. Baumgard. 2020. Effects of an oral supplement containing calcium and live yeast on post-absorptive metabolism, inflammation and production following intravenous lipopolysaccharide infusion in dairy cows. *Res. Vet. Sci.* 129:74–81. <https://doi.org/10.1016/j.rvsc.2020.01.007>.
- Allen, M. S. 1997. Relationship between fermentation acid production in the rumen and the requirement for physically effective fiber. *J. Dairy Sci.* 80:1447–1462. [https://doi.org/10.3168/jds.S0022-0302\(97\)76074-0](https://doi.org/10.3168/jds.S0022-0302(97)76074-0).
- AOAC. 1970. Official Methods of Analysis. 11th ed. Association of Official Analytical Chemists (AOAC).
- AOAC International. 2005. Official Methods of Analysis. 18th ed. AOAC International.
- Aslam, M. N., J. M. Kreider, T. Paruchuri, N. Bhagavathula, M. DaSilva, R. F. Zernicke, S. A. Goldstein, and J. Varani. 2010. A mineral-rich extract from the red marine algae *Lithothamnion calcareum* preserves bone structure and function in female mice on a Western-style diet. *Calif. Tissue Int.* 86:313–324. <https://doi.org/10.1007/s00223-010-9340-9>.
- Bernard, J. K., J. W. West, N. Mullis, Z. Wu, and S. J. Taylor. 2014. Evaluation of calcareous marine algae supplements on production and metabolic parameters of early lactation dairy cows. *Prof. Anim. Sci.* 30:649–656. <https://doi.org/10.15232/pas.2014-01339>.
- Birch, G. G., and O. M. Mwangiwa. 1974. Colorimetric determination of sugars in sweetened condensed milk products. *J. Sci. Food Agric.* 25:1355–1362. <https://doi.org/10.1002/jsfa.2740251103>.
- Bradford, B. J., K. Yuan, J. K. Farney, L. K. Mamedova, and A. J. Carpenter. 2015. Invited review: Inflammation during the transition to lactation: New adventures with an old flame. *J. Dairy Sci.* 98:6631–6650. <https://doi.org/10.3168/jds.2015-9683>.
- Cardoso, F. C., K. F. Kalscheur, and J. K. Drackley. 2020. Symposium review: Nutrition strategies for improved health, production, and fertility during the transition period. *J. Dairy Sci.* 103:5684–5693. <https://doi.org/10.3168/jds.2019-17271>.
- Cardoso, F. C., W. Sears, S. J. LeBlanc, and J. K. Drackley. 2011. Technical note: Comparison of 3 methods for analyzing areas under the curve for glucose and nonesterified fatty acids concentrations following epinephrine challenge in dairy cows. *J. Dairy Sci.* 94:6111–6115. <https://doi.org/10.3168/jds.2011-4627>.
- Cruywagen, C. W., S. Taylor, M. M. Beya, and T. Calitz. 2015. The effect of buffering dairy cow diets with limestone, calcareous marine algae, or sodium bicarbonate on ruminal pH profiles, production responses, and rumen fermentation. *J. Dairy Sci.* 98:5506–5514. <https://doi.org/10.3168/jds.2014-8875>.
- de Vries, M. J., and R. F. Veerkamp. 2000. Energy balance of dairy cattle in relation to milk production variables and fertility. *J. Dairy Sci.* 83:62–69. [https://doi.org/10.3168/jds.S0022-0302\(00\)74856-9](https://doi.org/10.3168/jds.S0022-0302(00)74856-9).
- Edmondson, A. J., I. J. Lean, L. D. Weaver, T. Farver, and G. Webster. 1989. A body condition scoring chart for Holstein dairy cows. *J. Dairy Sci.* 72:68–78. [https://doi.org/10.3168/jds.S0022-0302\(89\)79081-0](https://doi.org/10.3168/jds.S0022-0302(89)79081-0).
- Fuentes, M. C., S. Calsamiglia, P. Cardozo, and B. Vlaeminck. 2009. Effect of pH and level of concentrate in the diet on the production of biohydrogenation intermediates in a dual-flow continuous culture. *J. Dairy Sci.* 92:4456–4466. <https://doi.org/10.3168/jds.2008-1722>.
- Gaines, W. L. 1928. The energy basis of measuring milk yield in dairy cows. *Illinois Agric. Exp. Sta. Bull.* 308:403–436.
- Goff, J. P. 1999. Treatment of calcium, phosphorous, and magnesium balance disorders. *Vet. Clin. North Am. Food Anim. Pract.* 15:619–639. [https://doi.org/10.1016/S0749-0720\(15\)30167-5](https://doi.org/10.1016/S0749-0720(15)30167-5).
- Goff, J. P. 2008. The monitoring, prevention, and treatment of milk fever and subclinical hypocalcemia in dairy cows. *Vet. J.* 176:50–57. <https://doi.org/10.1016/j.tvjl.2007.12.020>.
- Goff, J. P., E. T. Littledike, and R. L. Horst. 1986. Effect of synthetic bovine parathyroid hormone in dairy cows: prevention of hypocalcemic parturient paresis. *J. Dairy Sci.* 69:2278–2289. [https://doi.org/10.3168/jds.S0022-0302\(86\)80666-X](https://doi.org/10.3168/jds.S0022-0302(86)80666-X).
- Gozho, G. N., D. O. Krause, and J. C. Plaizier. 2007. Ruminal lipopolysaccharide concentration and inflammatory response during grain-induced subacute ruminal acidosis in dairy cows. *J. Dairy Sci.* 90:856–866. [https://doi.org/10.3168/jds.S0022-0302\(07\)71569-2](https://doi.org/10.3168/jds.S0022-0302(07)71569-2).
- Grummer, R. R. 1995. Impact of changes in organic nutrient metabolism on feeding the transition dairy cow. *J. Anim. Sci.* 73:2820–2833. <https://doi.org/10.2527/1995.7392820x>.
- Heffernan, S. M., C. McCarthy, S. Eustace, R. E. FitzPatrick, E. Delahunty, and G. De Vito. 2020. Mineral rich algae with pine bark improved pain, physical function and analgesic use in mild-knee joint osteoarthritis, compared to glucosamine: A randomized controlled pilot trial. *Complement. Ther. Med.* 50:102349. <https://doi.org/10.1016/j.ctim.2020.102349>.
- Horst, E. A., S. K. Kvidera, and L. H. Baumgard. 2021. Invited review: The influence of immune activation on transition cow health and performance—A critical evaluation of traditional dogmas. *J. Dairy Sci.* 104:8380–8410. <https://doi.org/10.3168/jds.2021-20330>.
- Horst, E. A., E. J. Mayorga, M. Al-Qaisi, M. A. Abeyta, S. L. Portner, C. S. McCarthy, B. M. Goetz, S. K. Kvidera, and L. H. Baumgard. 2020. Effects of maintaining eucalcemia following immunoactiva-

- tion in lactating Holstein dairy cows. *J. Dairy Sci.* 103:7472–7486. <https://doi.org/10.3168/jds.2020-18268>.
- INRA. 2018. *Feeding Systems for Ruminants*. 1st ed. P. Nozière, D. Sauvant, and L. Delaby, ed. Wageningen Academic Publishers.
- Kawas, J. R., R. García-Castillo, H. Fimbres-Durazo, F. Garza-Cazares, J. F. G. Hernández-Vidal, E. Olivares-Sáenz, and C. D. Lu. 2007. Effects of sodium bicarbonate and yeast on nutrient intake, digestibility, and ruminal fermentation of light-weight lambs fed finishing diets. *Small Rumin. Res.* 67:149–156. <https://doi.org/10.1016/j.smallrumres.2005.09.010>.
- Kleen, J. L., G. A. Hooijer, J. Rehage, and J. P. T. M. Noordhuizen. 2003. Subacute ruminal acidosis (SARA): A review. *J. Vet. Med. A Physiol. Pathol. Clin. Med.* 50:406–414. <https://doi.org/10.1046/j.1439-0442.2003.00569.x>.
- Lancashire, P. D., H. Bleiholder, T. V. D. Boom, P. Langeluddeke, R. Stauss, E. Weber, and A. Witzemberger. 1991. A uniform decimal code for growth stages of crops and weeds. *Ann. Appl. Biol.* 119:561–601. <https://doi.org/10.1111/j.1744-7348.1991.tb04895.x>.
- Lean, I. J., P. DeGaris, D. McNeil, and E. Block. 2006. Hypocalcemia in dairy cows: Meta-analysis and dietary cation anion difference theory revisited. *J. Dairy Sci.* 89:669–684. [https://doi.org/10.3168/jds.S0022-0302\(06\)72130-0](https://doi.org/10.3168/jds.S0022-0302(06)72130-0).
- Leno, B. M., S. E. LaCount, C. M. Ryan, D. Briggs, M. Crombie, and T. R. Overton. 2017. The effect of source of supplemental dietary calcium and magnesium in the peripartum period, and level of dietary magnesium postpartum, on mineral status, performance, and energy metabolites in multiparous Holstein cows. *J. Dairy Sci.* 100:7183–7197. <https://doi.org/10.3168/jds.2017-12773>.
- Li, W., R. Angel, P. W. Plumstead, and H. Enting. 2021. Effects of limestone particle size, phytate, calcium source, and phytase on standardized ileal calcium and phosphorus digestibility in broilers. *Poult. Sci.* 100:900–909. <https://doi.org/10.1016/j.psj.2020.10.075>.
- Little, M. W., N. E. O'Connell, M. D. Welsh, F. J. Mulligan, and C. P. Ferris. 2017. Concentrate supplementation of a diet based on medium-quality grass silage for 4 weeks prepartum: Effects on cow performance, health, metabolic status, and immune function. *J. Dairy Sci.* 100:4457–4474. <https://doi.org/10.3168/jds.2016-11806>.
- Martinez, N., L. D. Sinedino, R. S. Bisinotto, E. S. Ribeiro, G. C. Gomes, F. S. Lima, L. F. Greco, C. A. Risco, K. N. Galvão, D. Taylor-Rodriguez, J. P. Driver, W. W. Thatcher, and J. E. P. Santos. 2014. Effect of induced subclinical hypocalcemia on physiological responses and neutrophil function in dairy cows. *J. Dairy Sci.* 97:874–887. <https://doi.org/10.3168/jds.2013-7408>.
- McArt, J. A. A., and R. C. Neves. 2020. Association of transient, persistent, or delayed subclinical hypocalcemia with early lactation disease, removal, and milk yield in Holstein cows. *J. Dairy Sci.* 103:690–701. <https://doi.org/10.3168/jds.2019-17191>.
- Mulligan, F. J., and M. L. Doherty. 2008. Production diseases of the transition cow. *Vet. J.* 176:3–9. <https://doi.org/10.1016/j.tvjl.2007.12.018>.
- Neville, E. W., A. G. Fahey, V. P. Gath, B. P. Molloy, S. J. Taylor, and F. J. Mulligan. 2019. The effect of calcareous marine algae, with or without marine magnesium oxide, and sodium bicarbonate on rumen pH and milk production in mid-lactation dairy cows. *J. Dairy Sci.* 102:8027–8039. <https://doi.org/10.3168/jds.2019-16244>.
- NRC. 2001. *Nutrient Requirements of Dairy Cattle*. 7th rev. ed. Natl. Acad. Press.
- O'Gorman, D. M., C. O'Carroll, and R. J. Carmody. 2012. Evidence that marine-derived, multi-mineral, Aquamin inhibits the NF- κ B signalling pathway *in vitro*. *Phytother. Res.* 26:630–632. <https://doi.org/10.1002/ptr.3601>.
- Penner, G. B., K. A. Beauchemin, and T. Mutsavangwa. 2007. Severity of ruminal acidosis in primiparous Holstein cows during the periparturient period. *J. Dairy Sci.* 90:365–375. [https://doi.org/10.3168/jds.S0022-0302\(07\)72638-3](https://doi.org/10.3168/jds.S0022-0302(07)72638-3).
- Quiroz-Rocha, G. F., S. J. LeBlanc, T. F. Duffield, D. Wood, K. E. Leslie, and R. M. Jacobs. 2009. Reference limits for biochemical and hematological analytes of dairy cows one week before and one week after parturition. *Can. Vet. J.* 50:383–388.
- Tyrrell, H. F., and J. T. Reid. 1965. Prediction of the energy value of cow's milk. *J. Dairy Sci.* 48:1215–1223. [https://doi.org/10.3168/jds.S0022-0302\(65\)88430-2](https://doi.org/10.3168/jds.S0022-0302(65)88430-2).
- Van Soest, P. J., J. B. Robertson, and B. A. Lewis. 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* 74:3583–3597. [https://doi.org/10.3168/jds.S0022-0302\(91\)78551-2](https://doi.org/10.3168/jds.S0022-0302(91)78551-2).
- Wu, Z., J. K. Bernard, and S. J. Taylor. 2015. Effect of feeding calcareous marine algae to Holstein cows prepartum or postpartum on serum metabolites and performance. *J. Dairy Sci.* 98:4629–4639. <https://doi.org/10.3168/jds.2014-8711>.
- Zadoks, J. C., T. T. Chang, and C. F. Konzak. 1974. A decimal code for the growth stages of cereals. *Weed Res.* 14:415–421. <https://doi.org/10.1111/j.1365-3180.1974.tb01084.x>.
- Zebeli, Q., B. U. Metzler-Zebeli, and B. N. Ametaj. 2012. Meta-analysis reveals threshold level of rapidly fermentable dietary concentrate that triggers systemic inflammation in cattle. *J. Dairy Sci.* 95:2662–2672. <https://doi.org/10.3168/jds.2011-5080>.
- Zenk, J. L., J. L. Frestedt, and M. A. Kuskowski. 2018. Effect of calcium derived from *Lithothamnion* sp. on markers of calcium metabolism in premenopausal women. *J. Med. Food* 21:154–158. <https://doi.org/10.1089/jmf.2017.0023>.

ORCIDS

A. G. Fahey  <https://orcid.org/0000-0002-4594-5767>
 F. J. Mulligan  <https://orcid.org/0000-0002-1787-0788>